Additional Problems and Exercises

SYSC 3120: Software Requirements Engineering

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Dr. Jason Jaskolka
Carleton University
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1. Consider an e-auction system on to support electronic bidding on items on the Internet from anywhere at any time. This system is made up of components such as sellers, buyers, shipping companies, an independent e-payment subsystem, e-mail systems, and the software to be developed or extended for inserting and advertising items, handling bids, billing highest bidders, recording evaluations of sellers and buyers, securing transactions and so forth. Global properties emerging from component interactions include the satisfaction of buyers getting wider access to interesting items, the satisfaction of sellers getting wider access to potential buyers, auction rules regulating the system, trustworthiness relationships and so on.
   (a) What is the **system-as-is?**
   (b) What is the **system-to-be**?
   (c) What is the **software-to-be**?

2. You are part of a team developing a smart traffic management system, designed to optimize traffic flow in a large metropolitan area. The system must comply with government regulations for traffic management and public safety. The system will use real-time data from sensors embedded in roads and traffic lights to monitor vehicle movement and adjust traffic signals accordingly. It must respond to changes in traffic conditions within 1 second to prevent congestion and accidents. The system should also be able to communicate with emergency vehicles, giving them priority at intersections.
   (a) Identify at least two **functional** requirements.
   (b) Identify at least two **nonfunctional** requirements.
   (c) Identify at least two **domain** requirements.

3. You are developing a comprehensive eHealth system designed to improve healthcare delivery by integrating patient data, telemedicine, and remote monitoring into a unified platform. The eHealth system is required to utilize Amazon Web Services (AWS) for cloud storage and computing, with a strict limit of 200ms for data retrieval from the cloud to ensure timely access to patient information, even during peak usage hours. The system must support real-time video consultations with a latency of no more than 50ms to ensure smooth and effective telemedicine sessions, even during periods of high demand. The system must achieve 99.999% uptime. This requires the implementation of redundant data centers across geographically distinct locations, with automatic failover capabilities and real-time data synchronization. The system needs to ensure that at least 99.95% of diagnostic data collected from remote monitoring devices is accurate and correctly attributed to the corresponding patient records. The system must seamlessly integrate with at least 90% of existing Electronic Health Record (EHR) systems used by local hospitals to enable the secure exchange of healthcare information across different platforms. The system's user interface be designed to accommodate users with varying levels of digital literacy, including a mandatory feature for voice navigation and multi-language support covering at least the five most commonly spoken languages. The system must comply with the Health Level Seven (HL7) Version 3 standards, particularly focusing on the accurate exchange of clinical documents. The system must implement automated data validation checks to ensure that no critical patient data is lost or corrupted during transfer between different healthcare providers. The system must also fully comply with the

Health Insurance Portability and Accountability Act (HIPAA), with encryption protocols like AES-256 for all patient data at rest and in transit, and mandatory quarterly security audits to identify and mitigate potential vulnerabilities. The system must employ a modular architecture, where each module must be independently upgradeable without causing downtime in other parts of the system. This includes a mandatory system for automated deployment of patches and updates, with detailed documentation and monitoring tools to enable quick identification and resolution of issues.

(a) Identify at least one **technology** constraint.

(b) Identify at least one **safety** constraint.

(c) Identify at least one **security** constraint.

(d) Identify at least one **reliability** constraint.

(e) Identify at least one **performance** constraint.

(f) Identify at least one **accuracy** constraint.

(g) Identify at least one **usability** constraint.

(h) Identify at least one **interoperability** constraint.

(i) Identify at least one **maintainability** constraint.

4. *Elevator Control System*

A popular hotel chain wants to develop an elevator system for a new hotel to be constructed. The hotel chain needs a software system for coordinating and moving the elevators between floors for the new hotel. The elevator and the electrical and mechanical control mechanisms are supplied by the elevator manufacturer. Each elevator has a set of buttons, one for each floor. These illuminate when pressed and cause the elevator to visit the corresponding floor. The illumination is cancelled when the corresponding floor is visited by the elevator. Each floor, except the first and last, has two buttons: one to request an up-elevator and one to request a down-elevator. These buttons illuminate when pressed. The illumination is cancelled when the elevator visits the floor and is either moving in the desired direction or has no outstanding requests. In the latter case, if both floor requests buttons are pressed, only one should be cancelled. Each elevator also has an emergency button which, when pressed causes a warning signal to be sent to the site manager. The elevator is then deemed "out of service" and shall not respond to any requests. Each elevator has a mechanism to cancel its "out of service" status.

(a) Write a **mission statement** for the *Elevator Control System*.

(b) Identify and represent the system boundaries using a **context diagram** for the *Elevator Control System*.

(c) Identify and prioritize a **set of stakeholders** for the *Elevator Control System*.

5. Which **elicitation techniques** would you use to elicit system requirements from the following stakeholder groups?

(a) Passengers for the *Train Control System*

(b) Meeting Initiators for the *Meeting Scheduler System*

(c) Library Administrators for the *Library Management Software*